universität
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Department of
Computer Science

Recommender Systems

Evaluation of Recommender Systems

Eva Zangerle

Foundations of Evaluation

Evaluation of RecSys

Framework for Evaluating RecSys (FEVR)

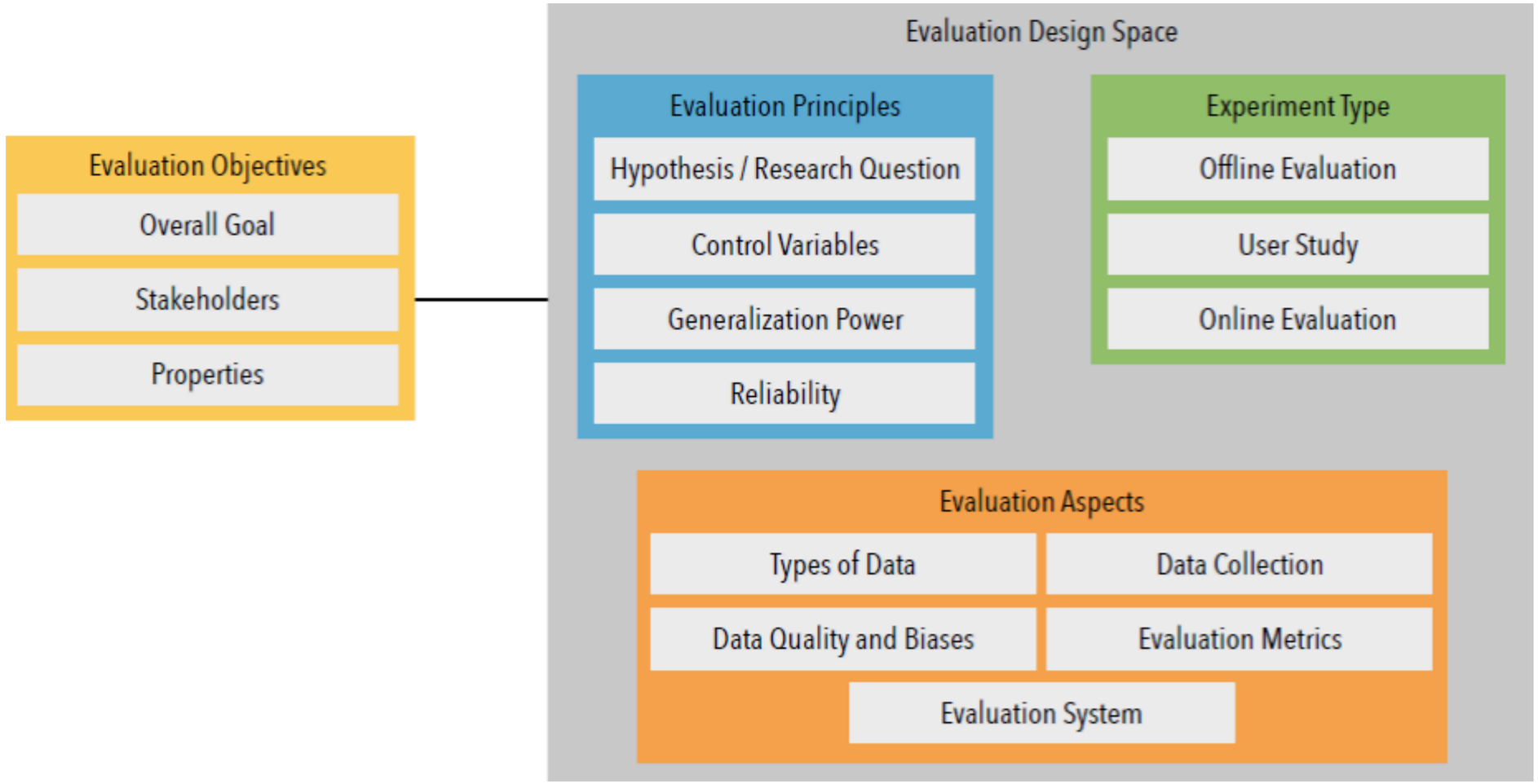


Figure taken from Eva Zangerle and Christine Bauer. 2022. Evaluating Recommender Systems: Survey and Framework. ACM Comput. Surv. 55, 8, Article 170 (August 2023). Doi: https://doi.org/10.1145/3556536

Evaluation Objectives

- Every evaluation should be goal-driven (e.g., increase user satisfaction, revenue, number of items sold, sell diverse items, help users understand the item space, etc.)
- Specific goal formulation: “Find the recommendation algorithm and parameterization that leads to the highest overall turnover on a specific ecommerce web site, if four product recommendations are displayed as a vertical list below the currently displayed product.” [Schröder 2011]
- Stakeholders
  - Objectives are shaped by the perspectives of stakeholders
  - Mostly, end user
  - But also: provider perspective, item provider perspective

Gunnar Schröder, Maik Thiele, and Wolfgang Lehner. 2011. Setting goals and choosing metrics for recommender system evaluations. In Joint proceedings of the ’11 Workshop on Human Decision Making in Recommender Systems (Decisions@RecSys’11) and User-Centric Evaluation of Recommender Systems and Their nterfaces-2 (UCERSTI 2) (Chicago, IL, USA) (UCERSTI 2), Vol. 811. CEUR-WS.org, Article 12.

Evaluation Objectives

- Properties
  - Identify properties that need to be improved (cf. overall goal)
  - Allows refined definition of goals

Evaluation Design Space

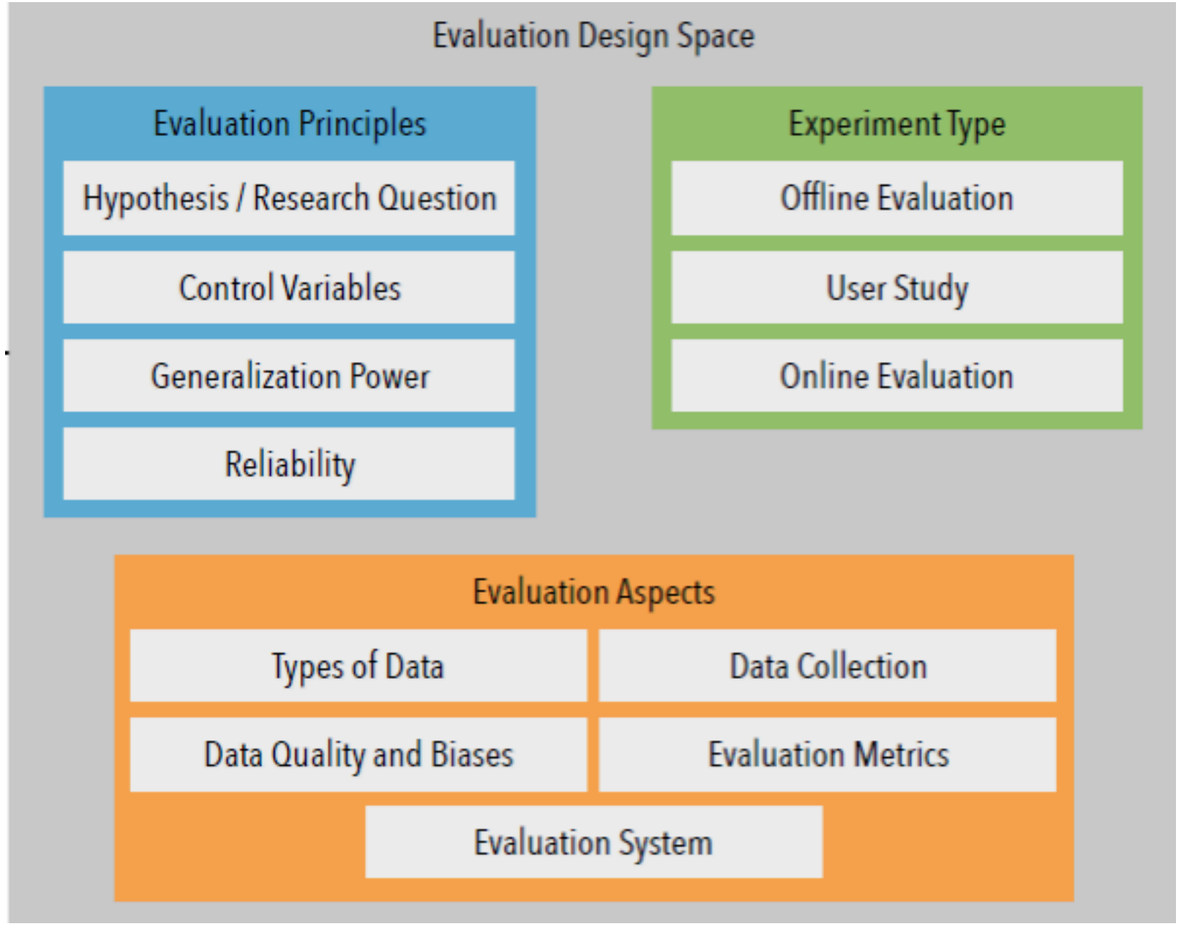
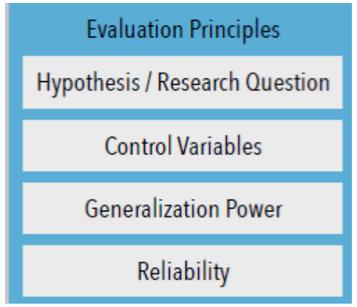


Figure taken from Eva Zangerle and Christine Bauer. 2022. Evaluating Recommender Systems: Survey and Framework. ACM Comput. Surv. 55, 8, Article 170 (August 2023). Doi: https://doi.org/10.1145/3556536

Experimental Design

Evaluation Principles



- Hypothesis
- Control Variables
- Generalization Power: To which extent are conclusions of the conducted evaluation generalizable beyond the current setup and experiments?
- Reliability
  - Consistency of multiple evaluation runs
  - Repeatability
  - Reproducibility

Definitions

- We use the Association for Computing Machinery (ACM)'s definitions.
- ACM definitions are aligned with the NISO RP-31-2021, Reproducibility Badging and Definitions document; approved by the (US) National Information Standards Organization (NISO).

- ACM uses these definitions for badges annotating papers:

Note

Note that in previous versions, ACM has used "swapped" definitions reproducibility and replication. Reasoning for the change: "As a result of discussions with the National Information Standards Organization (NISO), it was recommended that ACM harmonize its terminology and definitions with those used in the broader scientific research community, and ACM agreed with NISO's recommendation to swap the terms"reproducibility" and "replication" with the existing definitions used by ACM as part of its artifact review and badging initiative." (link).

Reproducibility

The measurement can be obtained with stated precision by a different team using the same measurement procedure, the same measuring system, under the same operating conditions, in the same or a different location on multiple trials. For computational experiments, this means that an independent group can obtain the same result using the author's own artifacts.
(Different team, same experimental setup)

Repeatability

The measurement can be obtained with stated precision by the same team using the same measurement procedure, the same measuring system, under the same operating conditions, in the same location on multiple trials. For computational experiments, this means that a researcher can reliably repeat her own computation. (Same team, same experimental setup)

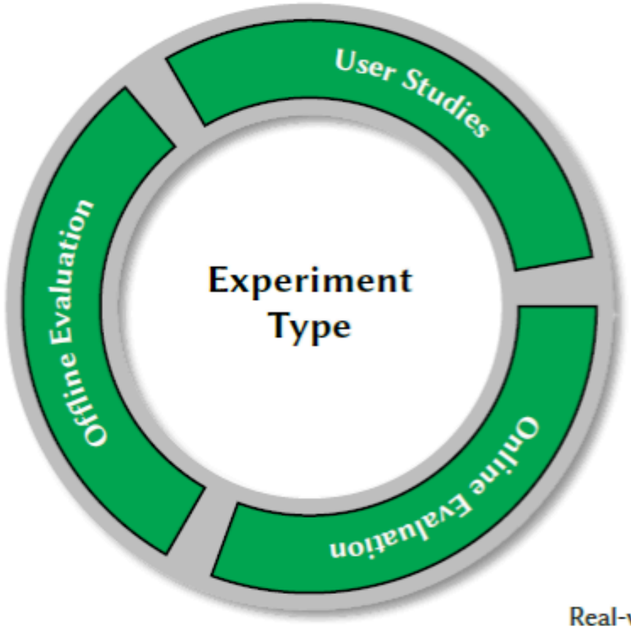
Replicability

The measurement can be obtained with stated precision by a different team, a different measuring system, in a different location on multiple trials. For computational experiments, this means that an independent group can obtain the same result using artifacts which they develop completely independently. (Different team, different experimental setup)

Experiment Type

Experiment Type
Offline Evaluation
User Study
Online Evaluation

User involved
Provided with tasks
Record interaction
Questionnaire



Historic data
Mimic user behavior
No user involved

Real-world settings
Productive system
Users involved

Figure taken from Eva Zangerle and Christine Bauer. 2022. Evaluating Recommender Systems: Survey and Framework. ACM Comput. Surv. 55, 8, Article 170 (August 2023). Doi: https://doi.org/10.1145/3556536

Experiment Type: Offline

- Compare predicted and known user-item-interactions
- Essentially, it’s simulation
- Predicted item is relevant if the user really consumed it
- Evaluation of retrieval task based on historic dataset

Cross Validation vs. Final Evaluation

Cross Validation vs. Final Evaluation cont.

Clean workflow:

1. Split data into training / validation / test
2. Use validation set or cross validation for model selection
3. Fix the final model and its hyperparameters
4. Retrain on the full training data
5. Evaluate once on the test set

> **Note**
>
> Do not use the test set for model selection.
> Otherwise, the reported performance estimate becomes optimistically biased.

Experimental Setup

- Model has to be evaluated on data it has not seen before ➜ cross-validation
- Two major categories:
  - Hold-out cross validation
  - k-fold cross validation

Experimental Setup: Hold-out Cross Validation

- Dataset is split into
  - Training data ➜ train model
  - Validation data ➜ used for model selection + tuning of hyperparameters
  - Test data ➜ estimate generalization power of model
- Hold-out method very sensitive to how the dataset is partitioned

Experimental Setup: Hold-out Cross Validation

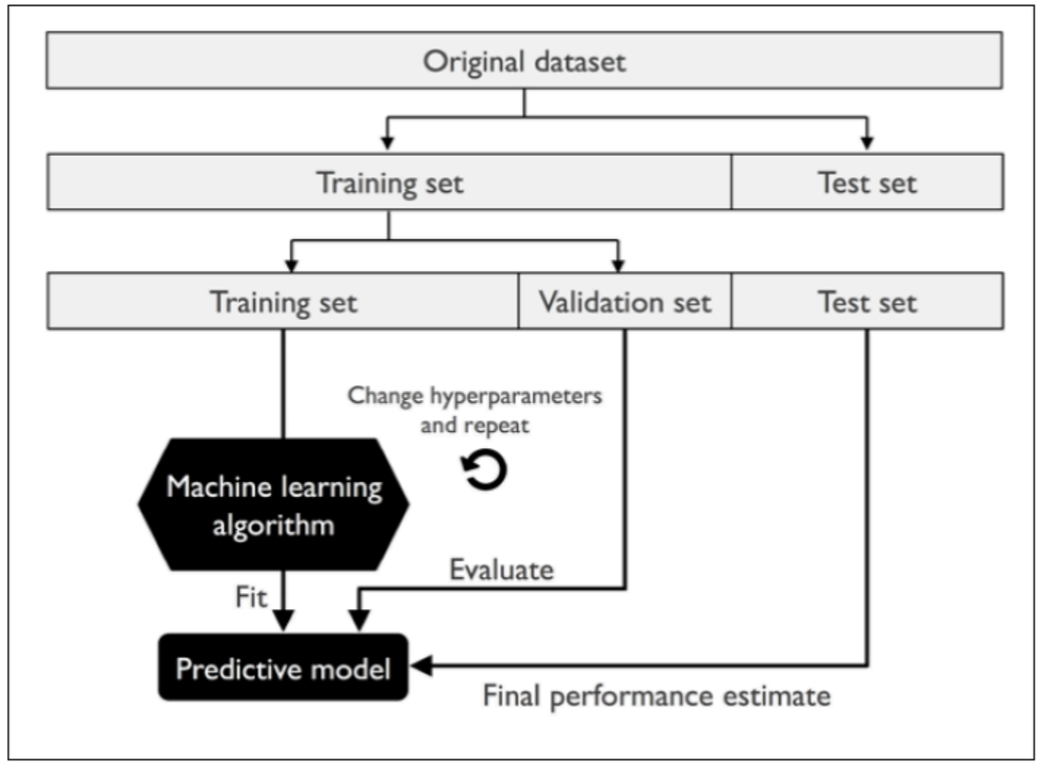


Figure taken from: Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow; Sebastian Raschka, Vahid Mirjalili; Packt Publishing Limited; 2nd edition, 2017.

Experimental Setup: k-Fold Cross Validation

- Split training set into k folds (without replacement)
  - k-1 folds used for training
  - 1 fold used for testing
- Repeat k times
- Every fold is used for test once, for training k-1 times
- Provides us with k models and k performance estimates
- More robust than hold-out cross validation

Experimental Setup: k-Fold Cross Validation

Figure taken from: Python Machine Learning: Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow; Sebastian Raschka, Vahid Mirjalili; Packt Publishing Limited; 2nd edition, 2017.

Experimental Setup: k-Fold Cross Validation

- Compute average performance of models based on folds
- Less sensitive to sub-partitioning of training data
- K-fold cross validation usually used for model tuning and hyperparameter tuning
- Once parameters are fixed:
  - Retrain model with fixed parameters on full training data
  - Apply model to test data
  - Compute final performance estimate

Experimental Setup: k-Fold Cross Validation

- Standard k-fold cross validation
  - Data directly divided into folds, or
  - Data first shuffled, then divided into folds (random)
- Stratified k-fold cross validation
  - Within each fold, class distribution of full training set is preserved
    ➜ Each fold representative of original class proportions

User-Centric Evaluation

- Problem with all quantitative effectiveness measures so far: Do they really assess if the recommended items satisfy the user?
- What does “satisfy” mean? (e.g., fulfilling an intent, same genre, suited for a specific situation, …)
- System can reach high accuracy just by making “easy” predictions (e.g., recommend always popular songs), but these may be useless for the user; novelty is sometimes defined as inverse of overall item popularity

User-Centric Evaluation

- Aspects to consider (“beyond accuracy”):
  - Similarity (items should match/be similar to the seed user’s taste)
  - Diversity (recommended items should not be too similar/boring)
  - Novelty / Familiarity (has the user already seen the item?)
  - Serendipity (user wants to discover something exciting, unexpected)
    - E.g., interesting item from another genre that the user usually does not like)
    - Hard to measure (contrasting accuracy)

User-Centric Evaluation cont.

- Explainability (recommender system should explain why an item was recommended); such as
  - List similar users and their tastes (“…because you friends like it.”)
  - Provide contextual explanations (“…because you listen to this kind of music at night.”)
  - Content-based explanations (“…because this movie features your favorite actor.”)

Experiment Type: User Study

- User observation in live or laboratory setting
- Users carry out a pre-defined task
- Users are observed
  - Log of user interactions
  - Questionnaire before/during/after task completion
  - Think aloud protocol
  - Eye tracking
  - Etc.

Experiment Type: Online Study

- Real-world user observation
- Task is selected by the user
- User is observed
  - Log of user interactions
  - Questionnaire before/during/after task completion

Offline vs. Online Evaluation

Offline evaluation

- Based on historical data
- Typical metrics:
  - RMSE
  - Precision@k / Recall@k
  - MAP, MRR, NDCG
- Fast, cheap, reproducible
- Useful for:
  - model development
  - model comparison
  - ablation studies

Online evaluation

- Based on real user interaction
- Typical metrics:
  - click-through rate
  - dwell time
  - conversion rate
  - retention
- Expensive, slower, riskier
- Useful for:
  - measuring actual user impact
  - validating business effects

Offline vs. Online Evaluation cont.

- Good offline performance does not guarantee good online performance
- Example:
  - Recommending very popular items may look good offline
  - But online, users may find the recommendations boring or unsurprising

Metrics as Proxies

Offline metrics are only proxies for user satisfaction.
Ultimately, only online evaluation can show whether a recommender system truly works in practice.

Experiment Type: Comparison

Type	Description
Offline	Method: simulation of user behavior based on past interactions Task: defined by researcher, purely algorithmic Repeatability: evaluation of arbitrary number of experiments (algorithmic settings, models, etc.) possible at low cost Scale: large dataset, large number of users Insights: quantitative, narrow (focused on predictive performance of algorithms)
User Study	Method: user observation in live or laboratory setting Task: defined by researcher, carried out by user Repeatability: expensive (recruitment of users) Scale: small cohort of users Insights: quantitative and/or qualitative (live user data, logging of user actions, eye tracking, questionnaires before/during/after task)
Online	Method: real-world user observation, online field experiment Task: self-selected by user, carried out by user Repeatability: expensive (requires full system and users) Scale: size of cohort of users depending on evaluation system and user base Insights: quantitative and/or qualitative (live user data, logging of user actions, questionnaires before/during/after exposure to system)

Table taken from Eva Zangerle and Christine Bauer. 2022. Evaluating Recommender Systems: Survey and Framework. ACM Comput. Surv. 55, 8, Article 170 (August 2023). Doi: https://doi.org/10.1145/3556536

Academic Practice

- Papers published 2017-2022 at major RecSys venues analyzed in a systematic literature study
- Initial set: 339 papers, reduced to 57 papers
- Goal: quantitative snapshot of the landscape of research on the evaluation of recommender systems
- Three key aspects: experiment type, datasets, and metrics.

Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170

Academic Practice

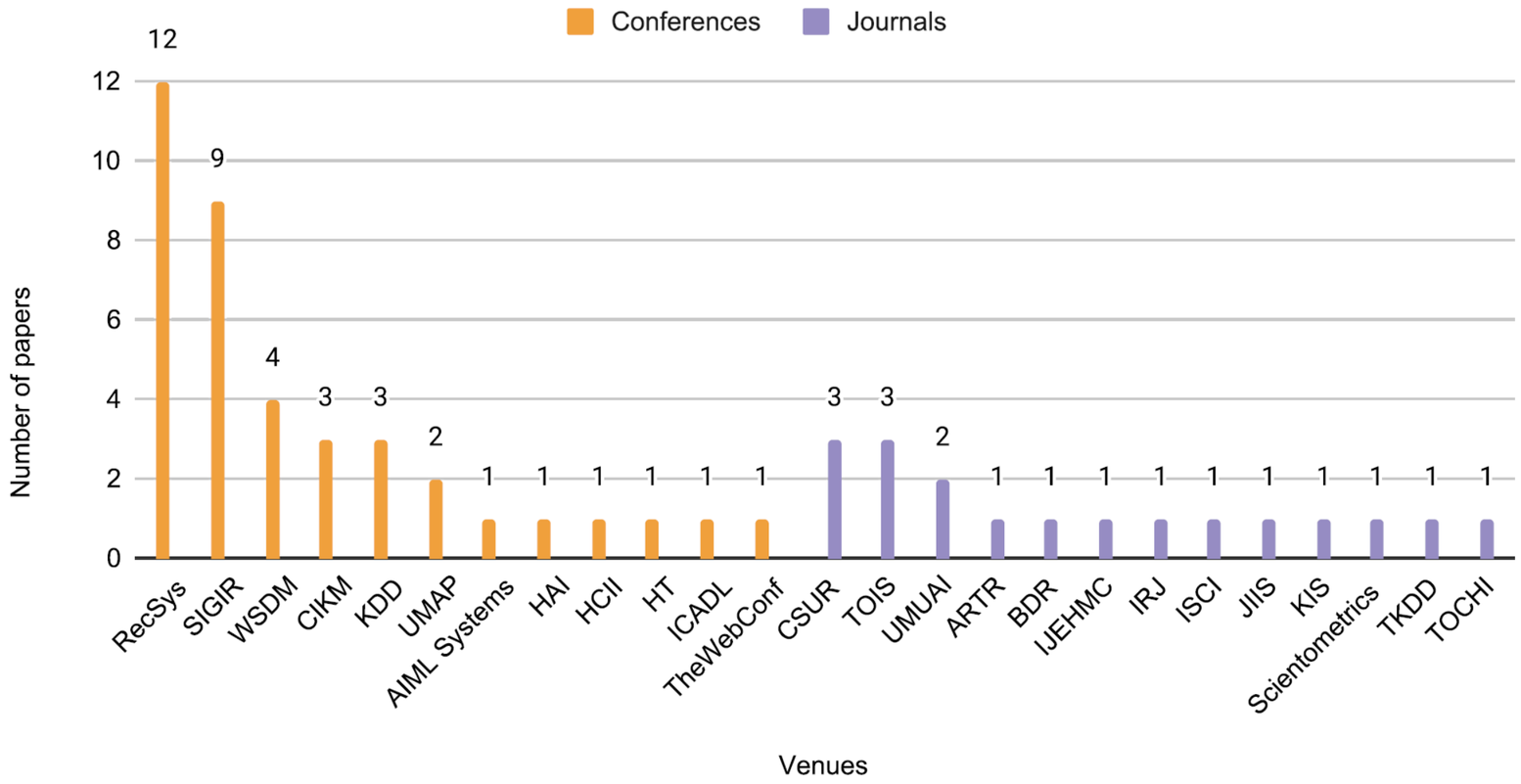


Fig. 2. Number of papers per venue, sorted by venue type (journals vs. conferences) and number of papers.

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Academic Practice

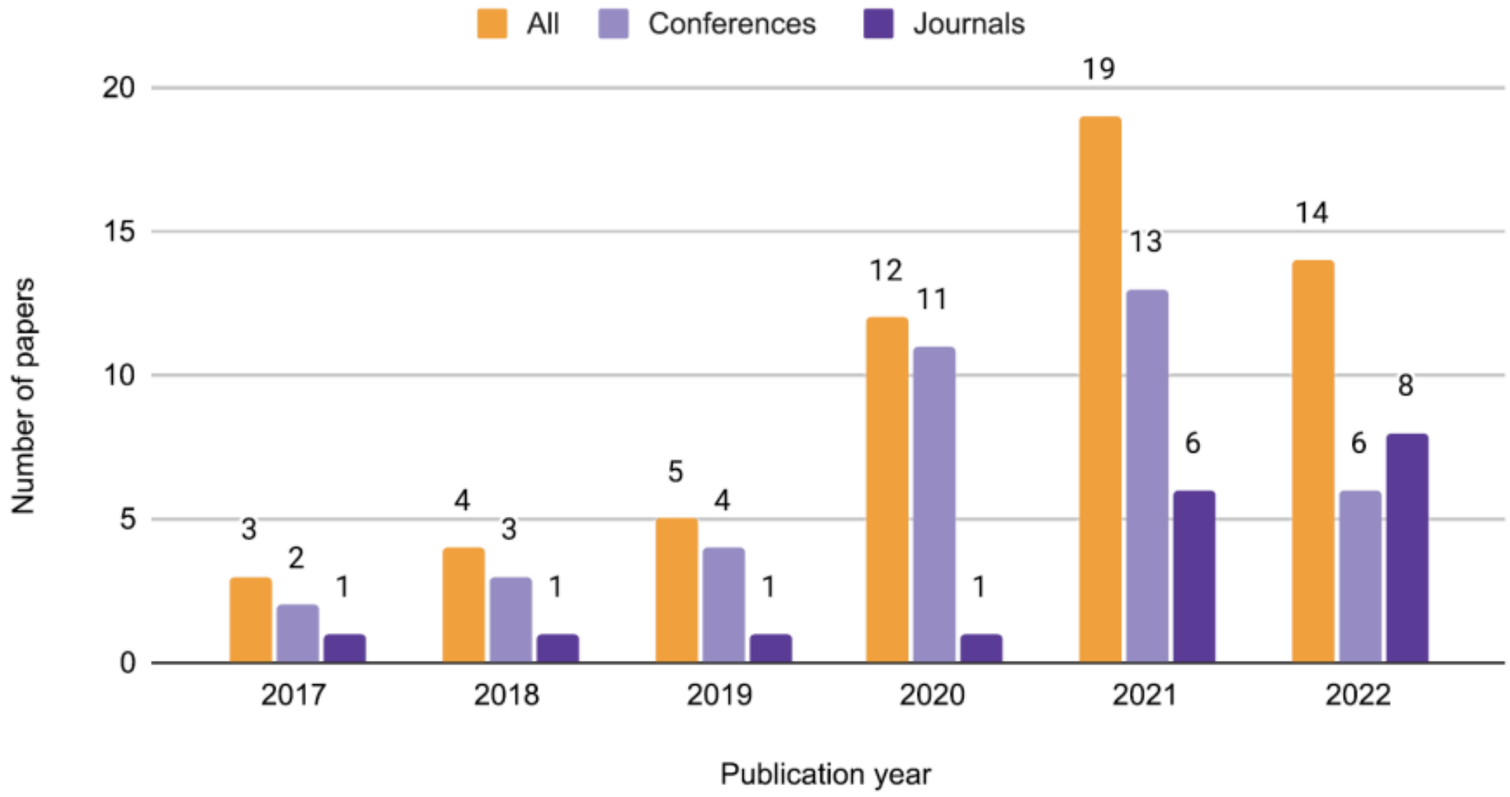

Fig. 3. Number of papers per year.

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Academic Practice

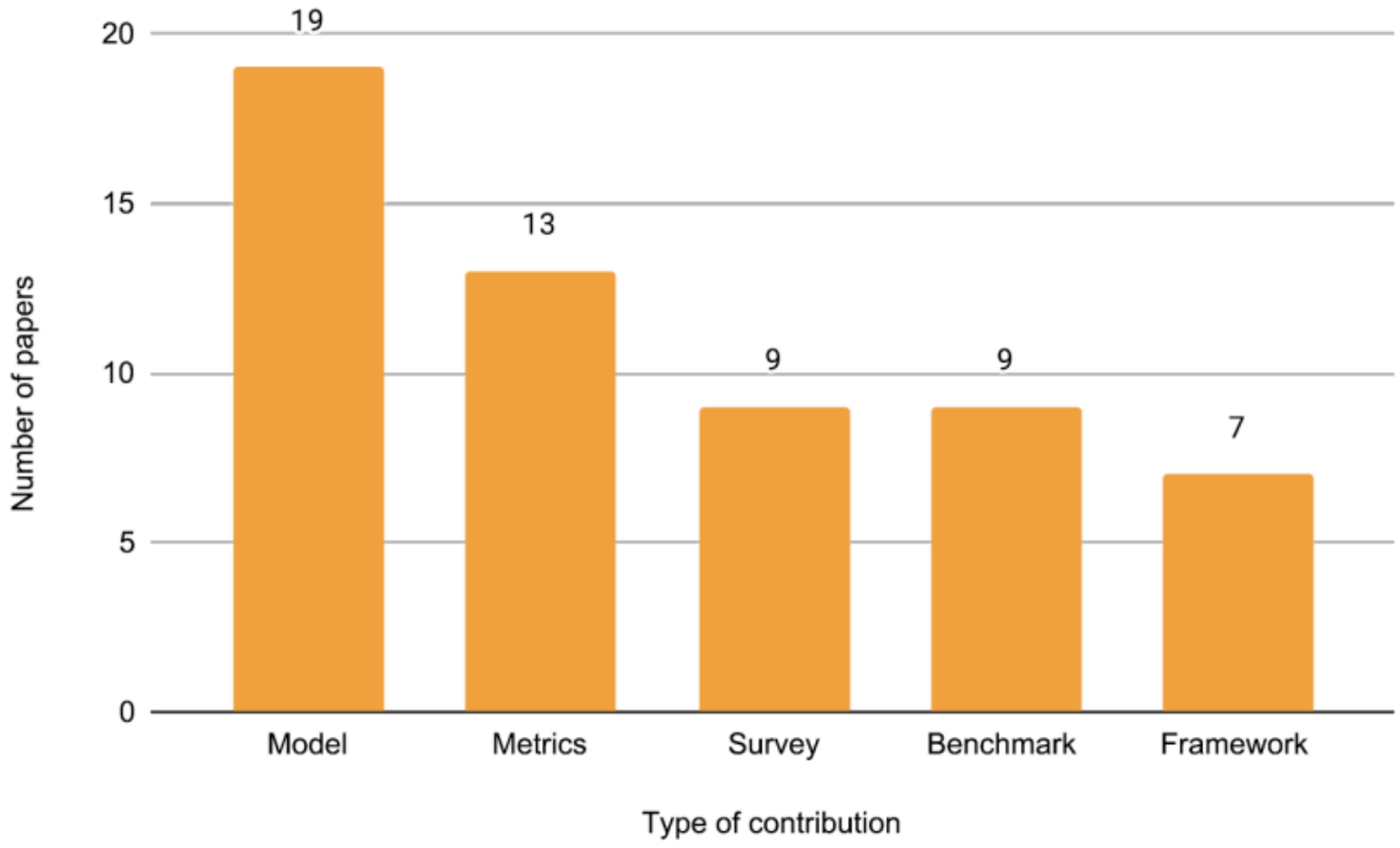

Fig. 4. Number of papers per type of contribution.

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Academic Practice: Experiment Type

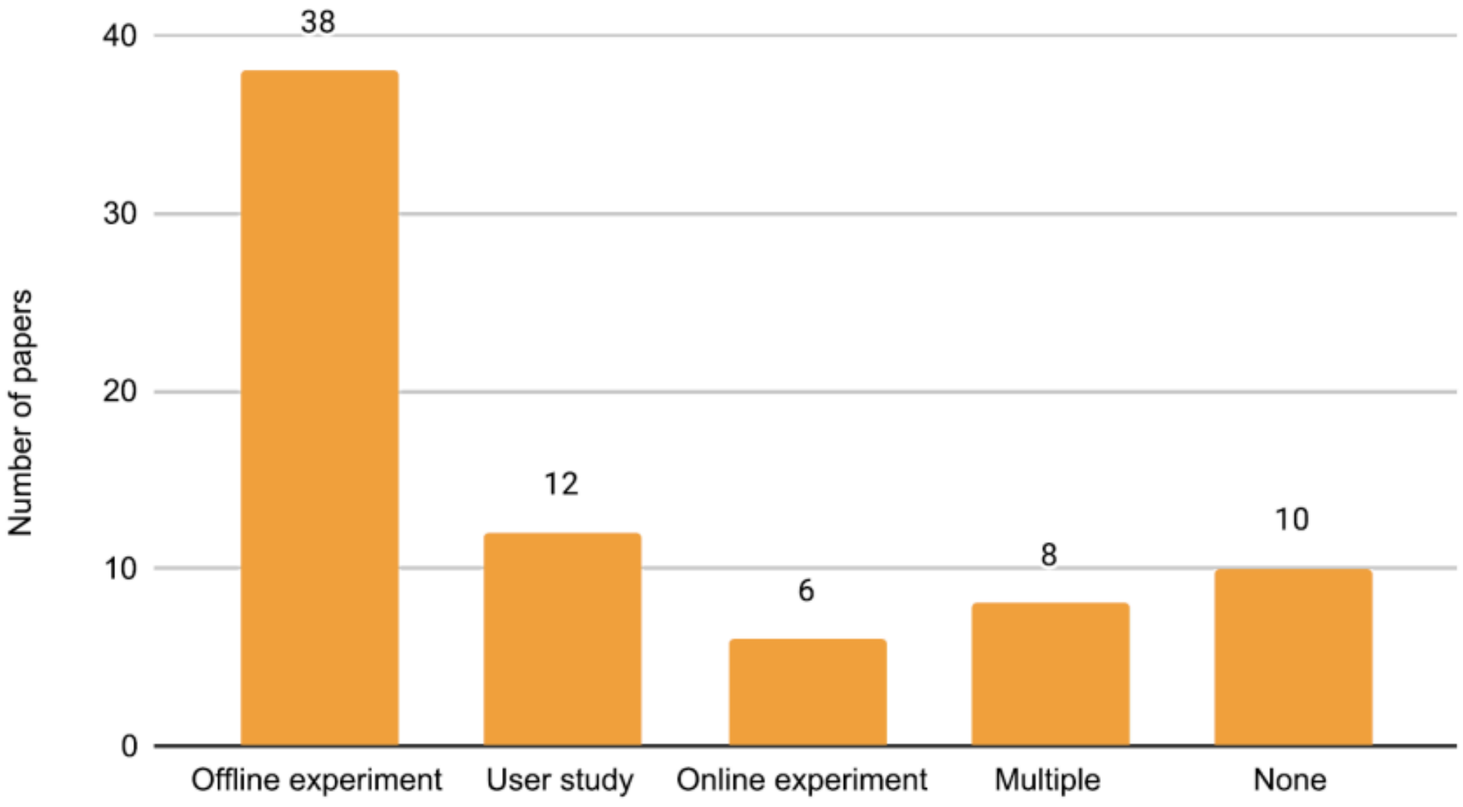

Fig. 5. Number of papers per experiment type.

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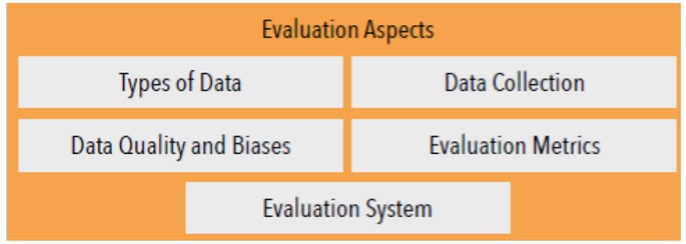
Academic Practice: Experiment Type

Table 5. Papers Using More Than One Experiment Type

Papers	Online experiment	Offline experiment	User study
Gilotte et al. [44]	x	x	
Narita et al. [73]	x	x	
Peska and Vojtas [77]	x	x	
Symeonidis et al. [91]	x	x	
Frumerman et al. [42]		x	x
Said and Bellogín [80]		x	x
Belavadi et al. [16]	x		x
Kouki et al. [59]	x	x	x

Figure taken from Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170

Evaluation Aspects



- Types of data
  - Implicit vs. explicit rating data

Dimension	Explicit Feedback	Implicit Feedback
Accuracy	High	Low
Abundance	Low	High
Expressivity of user preference	Positive and negative	Positive
Measurement reference	Absolute	Relative

  - User, item information (side information)
  - Qualitative vs. quantitative data vs. behavioral data
    - Quantitative: surveys, questionnaires
    - Qualitative: interviews, focus groups, participant observations
  - Natural vs. synthetic data

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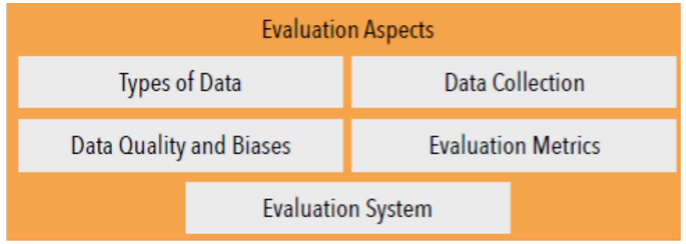
Evaluation Aspects cont.



- Data Collection
  - Existing datasets
  - Degree of user involvement
    - Recruit users for trial, allocate to RecSys and to baseline RecSys (control)
    - Field setting (A/B tests)
    - None (offline experiments)
  - User feedback elicitation
    - Interaction log
    - Observation
    - Self-reports (ask users for intentions in particular scenario)

Academic Practice: Datasets

Table 7. Details of Datasets Used in at Least Three Papers

Datasets	Domains	Feedback	Interactions	Side Information
Amazon Electronics, Products, Video Games [74]	Products	[1,5]	20,994,353 (E), 371,345 (B), 2,565,349 (V)	product information (e.g., description, color, product images, technical details), timestamp
citeulike-a, citeulike-t	Scientific Papers	{0,1}	204,987 (a), 134,860 (t)	tags, bag-of-words, and raw text for each article, citations between articles
epinions [92]	Products	[0,5]	922,267	explicit trust relationships among users, timestamps
LastFM [24]	Music	{0,1}	19,150,868	artist, song name, timestamp
MovieLens (100k, 1M, 20M) [48]	Movies	[0,5]	100,000 (100k)–20,000,000 (20M)	movie metadata (e.g., title, genre), user metadata (e.g., age, gender), rating timestamp
NetflixPrize[26]	Movies	[1,5]	100,000,000	movie metadata (title, release year), rating date
Yelp[27]	Business	[0,5]	6,990,280	business metadata (address, category, etc.), user metadata (user name, user stats (no. of reviews, user votes, etc.)), rating timestamp

We list the domain of the dataset, the type of feedback, number of interactions contained, and side information provided.

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Academic Practice: Datasets

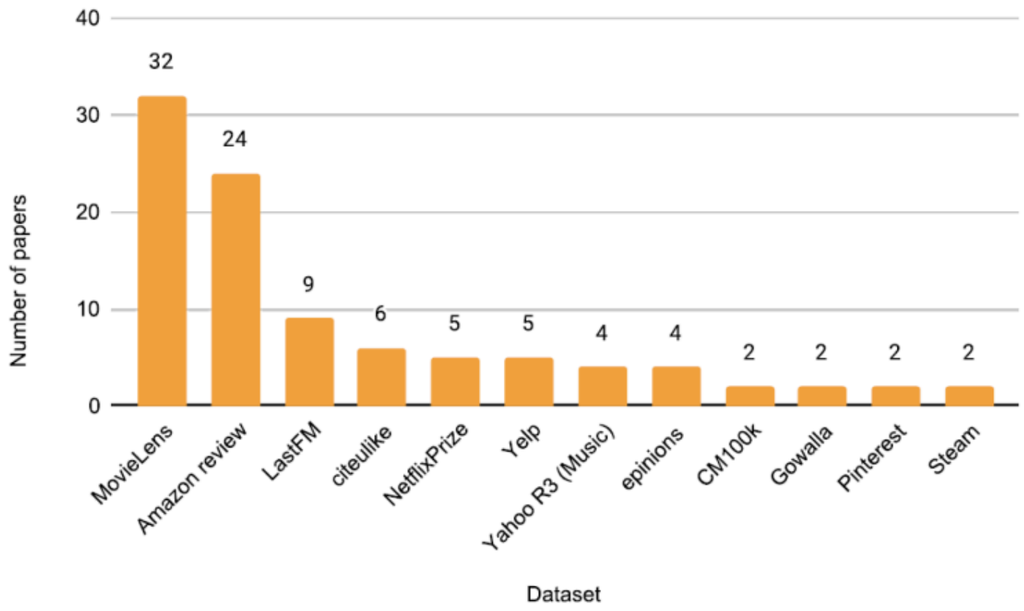

Fig. 6. Overview of datasets used in at least two papers, where different versions of a dataset are aggregated into a single dataset category for the Amazon review, MovieLens, and citeulike datasets.

Figure taken from Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170

Academic Practice: Datasets

Table 8. Combinations of Datasets (Pairs and Triples) Frequently Co-occurring in Experiments

Dataset Combinations	# Papers
{LastFM, ML 1M}	7
{ML 1M, NetflixPrize}, {ML 1M, Yelp}	5
{ML 1M, Yahoo R3}, {LastFM, Yelp}, {LastFM, NetflixPrize}, {LastFM, ML 1M, NetflixPrize}, {LastFM, ML 1M, Yelp}	4
{Amazon Movies & TV, LastFM}, {Amazon Electronics, LastFM}, {Amazon Beauty, ML 20M}, {epinions, ML 1M}, {ML 100k, ML 20M}, {ML 100k, ML 1M}, {ML 1M, ML20M}	3

We list all sets of datasets that co-occur in at least three papers (ML = MovieLens).

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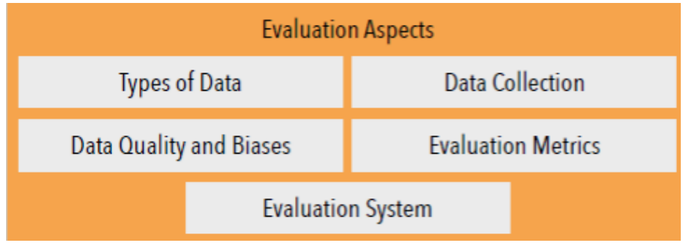
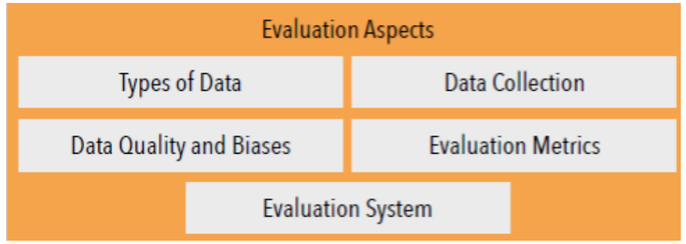
Evaluation Aspects



- Data quality and biases
- Evaluation system
  - No system (offline)
  - System dedicated to experimental use
  - Collaborate with operator of live systems
  - Develop and maintain a research system and its community

Evaluation Metrics

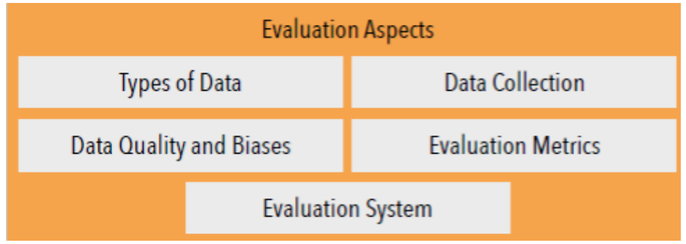
Evaluation Aspects



- Evaluation metrics
  - Variety of metrics available
  - Measure different aspects that are supposed to capture user experience and satisfaction (at least, to some extent)

Aspects: Metrics Overview

Category	Metrics
Prediction accuracy	mean absolute error (MAE) (root) mean squared error ((R)MSE)
Usage prediction	recall, precision, F-measure receiver operating characteristic curve (ROC) area under ROC curve (AUC)
Ranking	normalized discounted cumulative gain (NDCG) mean reciprocal rank (MRR)
Novelty	item novelty global long-tail novelty
Diversity	intra-list diversity
Coverage	item coverage user space coverage Gini index



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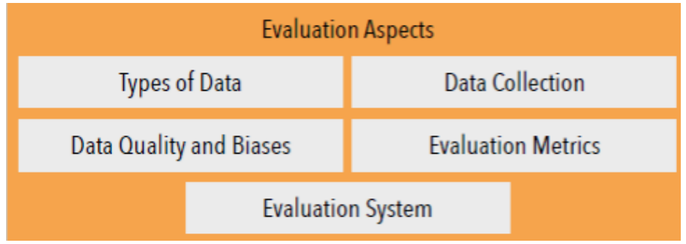
Aspects: Metrics Overview

Serendipity	unexpectedness serendipity
Fairness across users	value unfairness absolute unfairness over/underestimation of fairness
Fairness across items	pairwise fairness disparate treatment ratio (DTR) equal expected exposure equity of amortized attention disparate impact ratio (DIR) viable-Λ test
Business-oriented	click through rate (CTR) adoption and conversion rate sales and revenue



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Metrics: Why so many?

- Recommender systems can be evaluated from different perspectives:
  - Prediction: How close are predicted ratings to true ratings?
    $\longrightarrow$ MAE, RMSE
  - Retrieval: Did the system retrieve relevant items?
    $\longrightarrow$ Precision, Recall
  - Ranking: Are relevant items shown early in the list?
    $\longrightarrow$ MRR, MAP, NDCG
  - User experience: Are recommendations useful beyond accuracy?
    $\longrightarrow$ diversity, novelty, serendipity, coverage, explainability
- Different metrics reflect different goals, assumptions, and stakeholder perspectives (provider, user, item provider).

Metrics: Why so many? cont.

- There is no universally best metric. A metric is only meaningful if it matches the evaluation objective.
- Oftentimes, multiple metrics are used to capture different aspects of the system's performance. For example, a system might be evaluated for both accuracy (e.g., RMSE) and diversity (e.g., intra-list similarity).
- The choice of metric can significantly influence the conclusions drawn from an evaluation, and it is important to carefully select metrics that align with the specific goals of the recommender system being evaluated.

Metrics: Prediction Accuracy

- Evaluation under prediction aspects
- Predict ratings for unknown data items, evaluate “how far off” the predicted ratings are

Metrics: Mean Absolute Error

- Measure how close predicted ratings are to the true ratings.

$$MAE = \frac{1}{|T|} \sum_{(u,i) \in T} |r_{u,i} - \hat{r}_{u,i}|$$

where T is the test set, $r_{u,i}$ is the true rating of user u for item i, and $\hat{r}_{u,i}$ is the predicted rating.

  - Smaller MAE indicates better performance.
  - Sensitive to outliers (large errors have a bigger impact).

Metrics: Root Mean Squared Error

- De-facto in evaluating rating prediction accuracy (core task in early days of RecSys research).

$$RMSE = \sqrt{\frac{1}{|T|} \sum_{(u,i) \in T} (r_{u,i} - \hat{r}_{u,i})^2}$$

where T is the test set, $r_{u,i}$ is the true rating of user u for item i, and $\hat{r}_{u,i}$ is the predicted rating.

  - Smaller RMSE indicates better performance.
  - In contrast to MAE, RMSE penalizes larger errors more than smaller ones.
  - More sensitive to outliers than MAE.

Metrics: Usage Prediction

- Unranked retrieval: system returns a set of items without any particular order
- Evaluation based on set-based metrics
- Relevance judgments for items are binary (relevant vs. non-relevant)
- Evaluation based on confusion matrix:

	True Class: Positive	**True Class: Negative**
Predicted Class: Positive	**TP** (relevant & retrieved)	**FP** (not relevant but retrieved)
Predicted Class: Negative	**FN** (relevant but missed)	**TN** (not relevant & not retrieved)

Predicted relevant set: TP + FP
Actual relevant set: TP + FN

Metrics: Recall and Precision

- Input is set of recommended items, and set of relevant items.
- Recall: how many of the relevant items were recommended?

$$Recall = \frac{TP}{TP + FN} = \frac{|relevant\ items\ recommended|}{|all\ relevant\ items|}$$

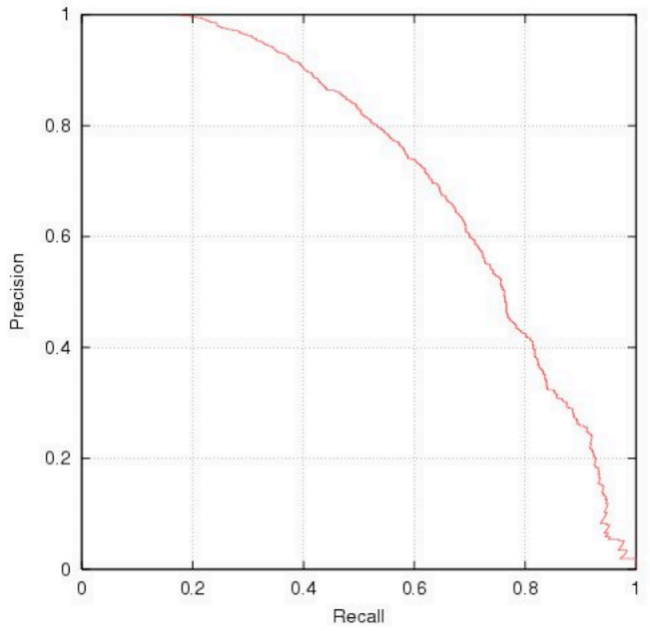
- Precision: how many of the recommended items were relevant?

$$Precision = \frac{TP}{TP + FP} = \frac{|relevant\ items\ recommended|}{|all\ items\ recommended|}$$

- Role of size of input size?
  - What if we recommend all items?

Metrics: Recall and Precision

- Recall and precision are inversely related
- Typically, when a recommender system is tuned to increase precision, recall decreases as a result (or vice versa)
- Optimize for recall or precision?



Metrics: Recall and Precision@k

- Assumption: user is in general not interested in all items, but only looks at a number of k highly ranked items.
- Recall@k: how many of the relevant items are in the top-k recommended items?

$$Recall@k = \frac{TP@k}{TP + FN} = \frac{|relevant\ items\ in\ top\text{-}k|}{|relevant\ items|}$$

- Precision@k: how many of the top-k recommended items are relevant?

$$Precision@k = \frac{TP@k}{TP@k + FP@k} = \frac{|relevant\ items\ in\ top\text{-}k|}{k}$$

Metrics: F-Score

- Combination of precision p and recall r into a single metric:

$$F_\beta = (1 + \beta^2) \cdot \frac{p \cdot r}{\beta^2 \cdot p + r}$$

- β parameter determines the weight of precision vs. recall in the combined score.
- $\beta = 1$: precision and recall are equally weighted, the metric is called F_1 score:

$$F_1 = 2 \cdot \frac{p \cdot r}{p + r}$$

- F1 score is the harmonic mean of precision and recall, which gives a more balanced measure when precision and recall are both important. It penalizes extreme values of either precision or recall, ensuring that a high F1 score can only be achieved when both precision and recall are reasonably high.

Metrics: Average Precision

- Average Precision (AP): average of precision values p at ranks where relevant items occur, where $rel_u(k)$ is an indicator function that is 1 if the item at rank k is relevant and 0 otherwise:

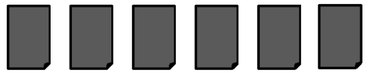
$$AP(u) = \frac{1}{|R_u|} \sum_{k=1}^{n} p_u@k \cdot rel_u(k)$$

  where $rel_u(k)$ is 1 if the item at rank k is relevant for user u, and 0 otherwise; R_u is the set of relevant items for user u.

- Mean Average Precision (MAP): mean of AP over all users U

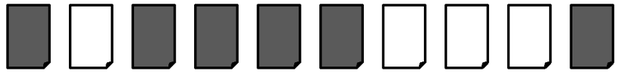
$$MAP = \frac{1}{|U|} \sum_{u \in U} AP(u)$$

Average Precision



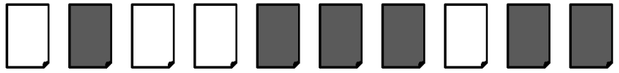
= the relevant documents

Ranking #1



Recall	0.17	0.17	0.33	0.5	0.67	0.83	0.83	0.83	0.83	1.0
Precision	1.0	0.5	0.67	0.75	0.8	0.83	0.71	0.63	0.56	0.6

Ranking #2



Recall	0.0	0.17	0.17	0.17	0.33	0.5	0.67	0.67	0.83	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.5	0.57	0.5	0.56	0.6

$$\text{Ranking \#1: } (1.0 + 0.67 + 0.75 + 0.8 + 0.83 + 0.6)/6 = 0.78$$

$$\text{Ranking \#2: } (0.5 + 0.4 + 0.5 + 0.57 + 0.56 + 0.6)/6 = 0.52$$

Mean Average Precision cont.

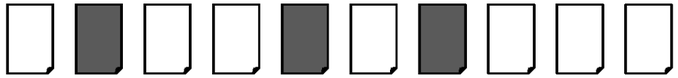
= relevant documents for query 1

Ranking #1

Recall	0.2	0.2	0.4	0.4	0.4	0.6	0.6	0.6	0.8	1.0
Precision	1.0	0.5	0.67	0.5	0.4	0.5	0.43	0.38	0.44	0.5

= relevant documents for query 2

Ranking #2



Recall	0.0	0.33	0.33	0.33	0.67	0.67	1.0	1.0	1.0	1.0
Precision	0.0	0.5	0.33	0.25	0.4	0.33	0.43	0.38	0.33	0.3

$$average\ precision\ query\ 1 = (1.0 + 0.67 + 0.5 + 0.44 + 0.5)/5 = 0.62$$
$$average\ precision\ query\ 2 = (0.5 + 0.4 + 0.43)/3 = 0.44$$

$$mean\ average\ precision = (0.62 + 0.44)/2 = 0.53$$

Figure taken from Manning, Raghavan, Schütze: Introduction to Information Retrieval, Cambridge University Press, 2008.

Metrics: Ranking

- Recall and precision primarily evaluate finding relevant items
- Rank metrics extend recall and precision to take the positions of correct items in a ranked list into account
- Evaluate finding and ranking
  - Relevant items are more useful when they appear earlier in the recommendation list
  - Particularly important in recommender systems as lower ranked items may be overlooked by users

Metrics: Reciprocal Rank

- So far, we have considered all relevant items equally important.
- Users are more likely to interact with items at the top of the list, so we want to give more weight to relevant items that appear earlier in the ranking.
- Assumption: user is satisfied with the first relevant item they encounter in the ranked list and stops looking further down the list.
- Reciprocal Rank (RR) is the inverse of the rank of the first relevant item in the ranked list for a given user u:

$$RR(u) = \frac{1}{rank_{u,i}}$$

where $rank_{u,i}$ is the rank of the first relevant item for user u in the ranked list of recommendations. If there are no relevant items in the list, RR is defined to be 0.

Metrics: Mean Reciprocal Rank (MRR)

- Mean Reciprocal Rank (MRR) is the average of the reciprocal ranks across all users

$$MRR = \frac{1}{|U|} \sum_{u \in U} RR(u)$$

where U is the set of users for whom we are evaluating the recommendations.

Metrics: Discounted Cumulative Gain (DCG)

- Result to a query item is an ordered list of items.
- Cumulative Gain (CG): graded relevance up to position k ("gain" for the user).

$$CG_k = \sum_{i=1}^{k} rel_i$$

where rel_i is the relevance score of the item at position i in the ranked list.

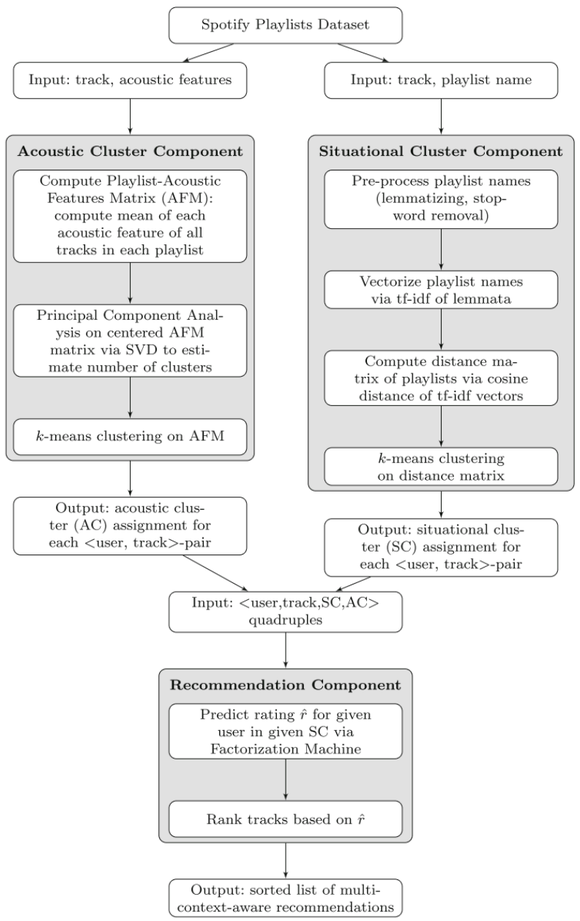
Metrics: Discounted Cumulative Gain (DCG)

- Example retrieved items: i1(rel=3), i2(rel=2), i3(rel=3), i4(rel=0), i5(rel=1), i6(rel=2), i7(rel=4 i8(rel=0).
- $CG@6 = \sum_{i=1}^{6} rel_i = 3 + 2 + 3 + 0 + 1 + 2 = 11$
- CG does not account for ordering of results, DCG does.
- $DCG@k = \sum_{i=1}^{k} \frac{rel_i}{\log_2(i+1)}$
- $DCG@6 = \frac{3}{\log_2(1+1)} + \frac{2}{\log_2(2+1)} + \frac{3}{\log_2(3+1)} + \frac{0}{\log_2(4+1)} + \frac{1}{\log_2(5+1)} + \frac{2}{\log_2(6+1)} \approx 8$
- Not normalized yet, so we cannot compare DCG scores across different queries with different numbers of relevant items.

Metrics: Example

- RecSys incorporating acoustic and situational features
- Evaluated using prediction accuracy and usage prediction metrics
- Input dataset (excerpt):

User	Track	SC	AC	AF_1	…	AF_7	Rating
872	309275	0	3	0.24		0.16	1
872	309275	1	3	0.24		0.16	0
872	250246	0	4	0.10		0.12	1
911	250246	2	4	0.10		0.12	1



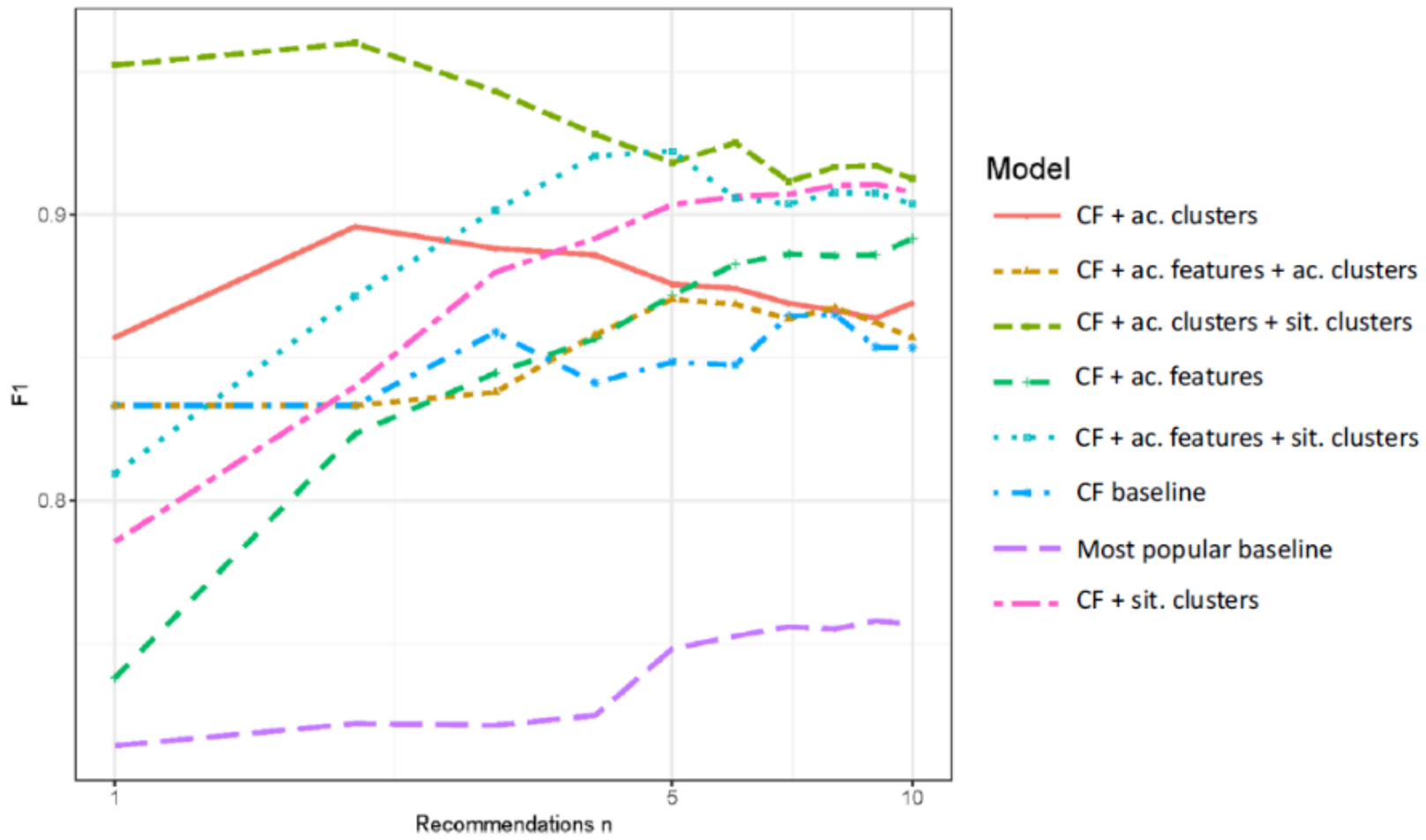
Pichl, M., Zangerle, E. User models for multi-context-aware music recommendation. Multimedia Tools and Applications 80, 22509–22531 (2021). https://doi.org/10.1007/s11042-020-09890-7

Metrics: Example cont.

Approach	Precision	Recall	F_1
AC+SC (CF + acoustic clusters + situational clusters)	**0.96**	0.81	**0.88**
AF+SC (CF + acoustic features + situational clusters)	0.80	0.78	0.79
AF (CF + acoustic features baseline)	0.86	0.68	0.76
AF+AC (CF + acoustic features + acoustic clusters)	0.85	0.65	0.74
AC (CF + acoustic clusters)	0.47	**0.85**	0.60
SC (CF + situational clusters)	0.46	0.83	0.59
CF (collaborative filtering baseline)	0.43	0.70	0.53
TR (theoretic random baseline)	0.50	0.50	0.50
MP (most popular baseline)	0.73	0.01	0.01

Pichl, M., Zangerle, E. User models for multi-context-aware music recommendation. Multimedia Tools and Applications 80, 22509–22531 (2021). https://doi.org/10.1007/s11042-020-09890-7

Metrics: Example



Pichl, M., Zangerle, E. User models for multi-context-aware music recommendation. Multimedia Tools and Applications 80, 22509–22531 (2021). https://doi.org/10.1007/s11042-020-09890-7

Metrics: Example

Approach	RMSE
AC+SC (CF + acoustic clusters + situational clusters)	**0.40**
AF+SC (CF + acoustic features + situational clusters)	**0.40**
AF (CF + acoustic features baseline)	0.44
AF+AC (CF + acoustic features + acoustic clusters)	0.47
AC (CF + acoustic clusters)	0.57
MP (most poplar baseline)	0.71
SC (CF + situational clusters)	0.72
CF (collaborative filtering baseline)	0.75

Pichl, M., Zangerle, E. User models for multi-context-aware music recommendation. Multimedia Tools and Applications 80, 22509–22531 (2021). https://doi.org/10.1007/s11042-020-09890-7

Academic Practice: Metrics

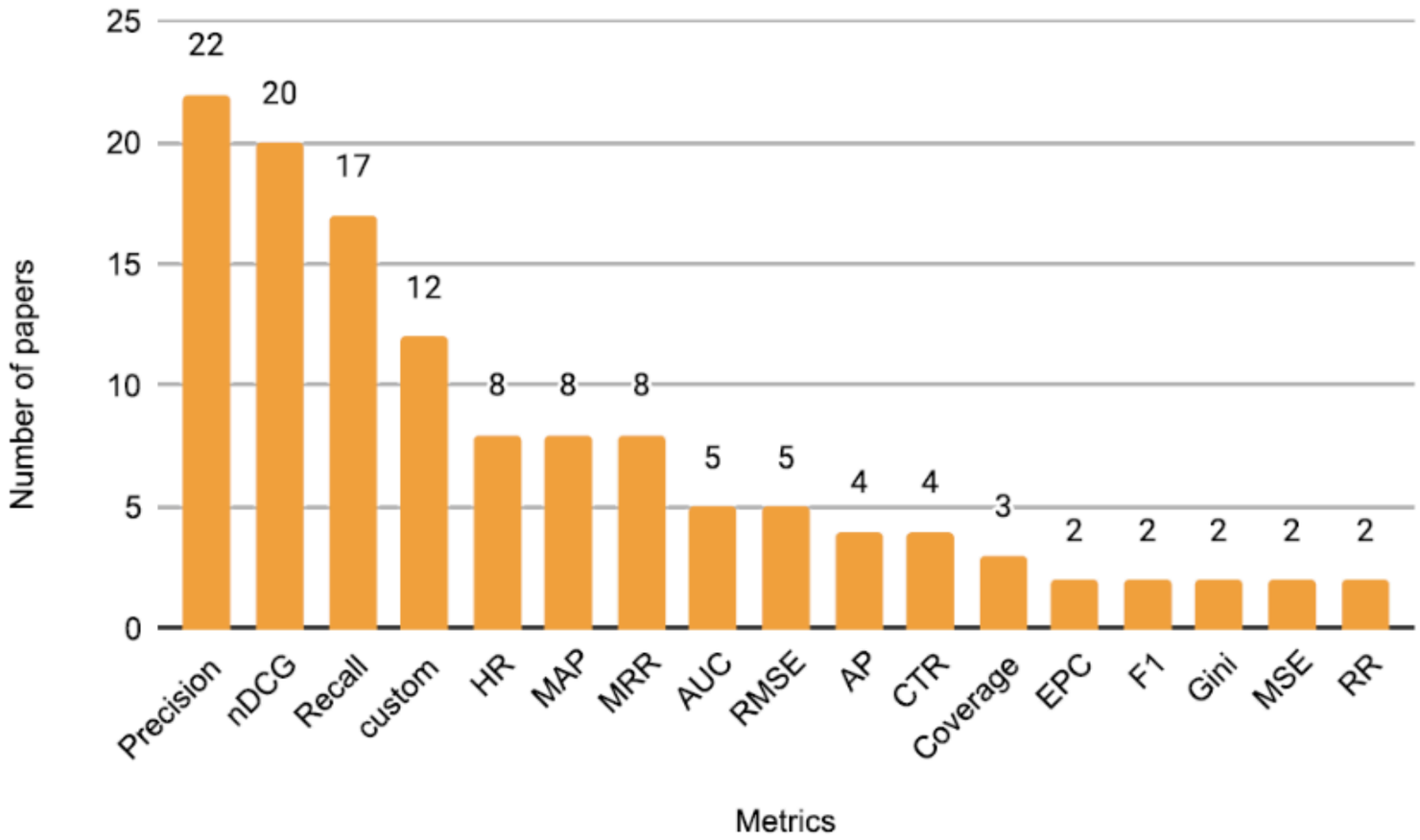


Fig. 7. Overview of metrics used in at least two papers (NB: Coverage refers to item coverage).

Figure taken from Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170

Academic Practice: Metrics

Table 10. The Categories of Value the Metrics Express

Categories	Metrics
Relevance	AP, AUC, F1, fallout, Hits, HR, InfAP, Logloss, MAP, P, R
Success Rate	CTR, CVR
Rating Prediction Accuracy	bpref, MAE, MSE, RMSE
Ranking	DCG, nDCG, MRR, RR
Non-accuracy	ACLT, APLT, Coverage, EFD, EPC, EPD, Gini, IC, ILD, Jaccard, Overlap, PCC, Popularity, PREO, PRSP

Table taken from Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170

Academic Practice: Metrics

Table 11. Combinations of Metrics Used Frequently in the Surveyed Papers

Metric combinations	# Papers
{nDCG, P}*	14
{nDCG, R}*	13
{P, R}	12
{nDCG, P, R}*	10
{nDCG, MAP}*, {R, MAP}, {nDCG, R, MAP}*	8
{nDCG, P, MAP}*, {P, MAP}, {nDCG, P, R, MAP}*, {nDCG, MRR}, {P, R, MAP}	7
{nDCG, MAP, MRR, R}*, {MRR, P, MAP, R}*, {nDCG, MRR, MAP}*, {MRR, MAP, R}*, {MRR, P, MAP}*, {nDCG, P, MRR, MAP}*, {MRR, P}*, {MRR, R}*, {MRR, MAP}*, {nDCG, HR}*, {nDCG, P, MRR, R}*, {MRR, HR}*, {MRR, P, R}*, {nDCG, P, MRR}*, {nDCG, MRR, R}*, {nDCG, MAP, MRR, P, R}*	6
{P, HR}, {nDCG, HR, MRR}*	5
{nDCG, P, HR, MAP}*, {P, HR, R, MAP}, {nDCG, HR, R}*, {nDCG, HR, R, MAP}*, {MRR, P, HR, R}*, {nDCG, P, HR, MRR}*, {nDCG, HR, MRR, R}*, {nDCG, P, HR, R}*, {MRR, MAP, HR, R}*, {MAP, MRR, P, HR, R}*, {nDCG, MRR, P, HR, MAP}*, {nDCG, MAP, MRR, HR, R}*, {nDCG, MAP, P, HR, R}*, {nDCG, MRR, P, HR, R}*, {nDCG, HR, MRR, MAP}*, {nDCG, MRR, P, HR, R, MAP}*, {MRR, P, HR, MAP}*, {nDCG, P, HR}*, {P, HR, R}, {MRR, HR, R}*, {MRR, P, HR}*, {nDCG, HR, MAP}*, {HR, R, MAP}, {P, HR, MAP}, {MRR, HR, MAP}*, {HR, R}, {HR, MAP}	4
{Coverage, HR}*, {P, AUC}, {AUC, R}, {nDCG, AUC}*, {P, Coverage, HR}*, {P, Coverage}*, {nDCG, AUC, R}*, {nDCG, AP}*, {AP, R}	3

Tuples with asterisks contain metrics from at least two of the categories in Table 10, excluding custom metrics. (NB: Coverage in refers to item coverage).

Table taken from Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170

Use Case Example

- Academic setting
- Novel recommendation algorithm: RecAlg
- Help users find likable music by personalized music recommendations
- Improve diversity of music recommendation by incorporating lyrics features of tracks
- Also: improve (at least, maintain) prediction accuracy

Use Case Example cont.

FEVR Component	Outcome
Evaluation Objectives	
Overall Goal	To evaluate whether users are able to find likable music in the recommendations computed by the novel RecAlg algorithm
Stakeholders	Users of the system (algorithm) Artists may also benefit from an increased item diversity as a more diverse set of artists may be represented
Properties	Item diversity in the recommendations; catalog coverage
Evaluation Principles	
Hypothesis / Research Question	H_1: The novel RecAlg approach provides users (on average) with more diverse recommendations with respect to the intra-list diversity while maintaining prediction accuracy compared to the baseline algorithm
Control Variables	Follow accountability framework by Bellogín and Said [25] (for randomization of users in dataset splitting to prevent selection bias)
Generalization Power	Limited due to lack of user involvement and dataset biases
Reliability	Follow accountability framework by Bellogín and Said [25]
Experiment Type	Offline Evaluation with A/B-testing
Evaluation Aspects	
Types of Data	Implicit ratings (listening events), side information for music tracks
Data Collection	LFM-2b dataset [192]
Data Quality and Biases	Platform bias, popularity bias, skewed gender distribution, imbalanced country distribution.
Evaluation Metrics	Prediction accuracy with RMSE; intra-list diversity in terms of different unique artists
Evaluation System	Existing evaluation framework ELLIOT [14]

Table taken from Christine Bauer, Eva Zangerle, and Alan Said. 2024. Exploring the Landscape of Recommender Systems Evaluation: Practices and Perspectives. ACM Transactions on Recommender Systems 2, 1, Article 11 (March 2024), 31 pages. https://doi.org/10.1145/3629170